

Algebra II with Trigonometry

Chapter 11.3

Olympiad Solutions (Gemini)

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Solutions

1. Let P be an arbitrary point on the hyperbola $16x^2 - 9y^2 = 144$. Find the product of the perpendicular distances from P to the two asymptotes of the hyperbola.

Solution: Dividing the hyperbola's equation by 144 yields its standard form:

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

The asymptotes are found by setting the constant to 0, yielding $16x^2 - 9y^2 = 0$, which cleanly factors into the two lines $4x - 3y = 0$ and $4x + 3y = 0$.

Let $P(x_0, y_0)$ be an arbitrary point on the hyperbola, so we know $16x_0^2 - 9y_0^2 = 144$. The perpendicular distances from P to the two asymptotes are given by the point-to-line distance formula:

$$d_1 = \frac{|4x_0 - 3y_0|}{\sqrt{4^2 + (-3)^2}} = \frac{|4x_0 - 3y_0|}{5}, \quad d_2 = \frac{|4x_0 + 3y_0|}{\sqrt{4^2 + 3^2}} = \frac{|4x_0 + 3y_0|}{5}$$

Taking the product of these distances:

$$d_1 d_2 = \frac{|4x_0 - 3y_0| \cdot |4x_0 + 3y_0|}{25} = \frac{|16x_0^2 - 9y_0^2|}{25}$$

Since P is on the hyperbola, the numerator is exactly 144. Thus, the product of the distances is intrinsically constant and equals $\frac{144}{25}$.

2. Point P lies on the hyperbola $\frac{x^2}{25} - \frac{y^2}{144} = 1$. If F_1 and F_2 are the foci of the hyperbola and the angle $\angle F_1PF_2$ is exactly 90° , determine the area of $\triangle F_1PF_2$.

Solution: From the equation, $a^2 = 25$ and $b^2 = 144$. The focal distance c is given by $c^2 = a^2 + b^2 = 169$, meaning $c = 13$. The distance between the foci is $F_1F_2 = 2c = 26$.

By the definition of a hyperbola, the absolute difference of the distances from any point on it to the foci is $2a$:

$$|PF_1 - PF_2| = 2a = 10$$

Squaring both sides of this equation yields:

$$PF_1^2 + PF_2^2 - 2PF_1 \cdot PF_2 = 100$$

Since $\angle F_1PF_2 = 90^\circ$, $\triangle F_1PF_2$ is a right triangle. By the Pythagorean theorem, the sum of the squares of its legs equals the square of the hypotenuse:

$$PF_1^2 + PF_2^2 = F_1F_2^2 = 26^2 = 676$$

Substitute this directly into our squared equation:

$$676 - 2PF_1 \cdot PF_2 = 100 \implies 2PF_1 \cdot PF_2 = 576$$

The area of right-angled $\triangle F_1PF_2$ is simply $\frac{1}{2}PF_1 \cdot PF_2$. Therefore, the area is $\frac{576}{4} = \mathbf{144}$. (Remark: As a general theorem, the area of such a focal right triangle is always b^2 .)

3. An ellipse and a hyperbola share the same foci, $F_1(-5, 0)$ and $F_2(5, 0)$. The ellipse has a major axis of length 14, and the hyperbola has a transverse

axis of length 8. If the two curves intersect at a point P in the first quadrant, what is the straight-line distance from P to the origin?

Solution: Let $d_1 = PF_1$ and $d_2 = PF_2$. Because P lies on the ellipse with a major axis of 14, the sum of its focal distances is:

$$d_1 + d_2 = 14$$

Because P lies on the hyperbola with a transverse axis of 8, and is in the first quadrant (placing it closer to F_2 than F_1), the difference of its focal distances is positive:

$$d_1 - d_2 = 8$$

Adding these equations yields $2d_1 = 22 \implies d_1 = 11$, and subtracting them gives $2d_2 = 6 \implies d_2 = 3$.

We need the distance PO from P to the origin O . Because the origin is the midpoint of F_1F_2 , segment PO is the median to side F_1F_2 in $\triangle PF_1F_2$. We apply Apollonius's Theorem:

$$PF_1^2 + PF_2^2 = 2(PO^2 + OF_2^2)$$

Substituting our known lengths ($OF_2 = 5$):

$$11^2 + 3^2 = 2(PO^2 + 5^2)$$

$$121 + 9 = 2PO^2 + 50 \implies 130 = 2PO^2 + 50$$

$$2PO^2 = 80 \implies PO^2 = 40$$

Thus, the straight-line distance to the origin is $PO = \sqrt{40} = 2\sqrt{10}$.

4. A circle centered at the origin intersects the hyperbola $x^2 - y^2 = 7$ at four distinct points, which form the vertices of a rectangle. If the area of this rectangle is 48, find the radius of the circle.

Solution: Let the coordinates of the first-quadrant vertex of the rectangle be (x_0, y_0) with $x_0, y_0 > 0$. By the combined symmetry of the origin-centered

circle and hyperbola, the four vertices are $(\pm x_0, \pm y_0)$. The side lengths of this rectangle are $2x_0$ and $2y_0$. We are given its area:

$$(2x_0)(2y_0) = 48 \implies 4x_0y_0 = 48 \implies 2x_0y_0 = 24$$

Because (x_0, y_0) is on the hyperbola, we have $x_0^2 - y_0^2 = 7$. The radius r of the circle is the distance from the origin to (x_0, y_0) , meaning $r^2 = x_0^2 + y_0^2$. Instead of solving for x_0 and y_0 individually, we make a quick algebraic observation to relate r^2 directly to our equations:

$$(r^2)^2 = (x_0^2 + y_0^2)^2 = (x_0^2 - y_0^2)^2 + (2x_0y_0)^2$$

Substitute the known values:

$$r^4 = 7^2 + 24^2 = 49 + 576 = 625$$

Taking the square root gives $r^2 = 25$. Because the radius must be positive, $r = 5$.

5. A hyperbola centered at the origin has its foci on the x -axis. Its asymptotes intersect at the origin to form an angle of 60° that contains the transverse axis. If the distance between the two vertices is 12, find the distance between the two foci.

Solution: Let the standard equation of the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The distance between the vertices is $2a = 12$, meaning $a = 6$.

The asymptotes are $y = \pm \frac{b}{a}x$. Because the angle between them (containing the transverse x -axis) is 60° , the symmetry implies that the angle between each asymptote and the positive x -axis is exactly half of that, which is 30° . The slope of the asymptote is $\tan(30^\circ)$, hence:

$$\frac{b}{a} = \frac{1}{\sqrt{3}} \implies b = \frac{a}{\sqrt{3}} = \frac{6}{\sqrt{3}} = 2\sqrt{3}$$

For any hyperbola, the focal length c satisfies $c^2 = a^2 + b^2$. Substituting our values for a and b :

$$c^2 = 6^2 + (2\sqrt{3})^2 = 36 + 12 = 48 \implies c = \sqrt{48} = 4\sqrt{3}$$

The distance between the two foci is $2c$, which equals $8\sqrt{3}$.

6. A line tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at a point P in the first quadrant intersects the hyperbola's asymptotes at points Q and R . Find the area of the triangle formed by Q , R , and the origin.

Solution: A beautiful classical property of hyperbolas is that the triangle formed by *any* tangent line and the two asymptotes bounds a constant area equal to ab . Let's prove it concisely:

For a generic hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the tangent at point $P(x_0, y_0)$ is $\frac{x_0x}{a^2} - \frac{y_0y}{b^2} = 1$. The asymptotes are $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$. Intersecting the tangent with $y = \frac{b}{a}x$ yields point Q :

$$x_Q \left(\frac{x_0}{a^2} - \frac{y_0}{ab} \right) = 1 \implies x_Q = \frac{a^2b}{bx_0 - ay_0}, \quad y_Q = \frac{ab^2}{bx_0 - ay_0}$$

Intersecting the tangent with $y = -\frac{b}{a}x$ yields point R :

$$x_R \left(\frac{x_0}{a^2} + \frac{y_0}{ab} \right) = 1 \implies x_R = \frac{a^2b}{bx_0 + ay_0}, \quad y_R = \frac{-ab^2}{bx_0 + ay_0}$$

Using the determinant formula for the area of $\triangle OQR$ from the origin:

$$\text{Area} = \frac{1}{2} |x_Q y_R - x_R y_Q| = \frac{1}{2} \left| \frac{-2a^3b^3}{(bx_0)^2 - (ay_0)^2} \right| = \frac{a^3b^3}{a^2b^2 \left(\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} \right)}$$

Because $P(x_0, y_0)$ resides on the hyperbola, the expression $\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} = 1$. The denominator simplifies to a^2b^2 , leaving exactly:

$$\text{Area} = \frac{a^3b^3}{a^2b^2} = ab$$

For the given hyperbola, $a^2 = 16 \implies a = 4$ and $b^2 = 9 \implies b = 3$. The area of the triangle is simply $4 \times 3 = \mathbf{12}$.